

Credit Scoring Model Using Machine Learning Methods

Credit Risk Modelling — Semester Project | **Group 1**

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Dataset	German Credit Data (Statlog) — UCI ML Repository
Records	1,000 loan applicants, 20 features
Task	Binary classification: predict probability of default
Models	Logistic Regression, LDA, Decision Tree, Random Forest, Gradient Boosting
Software	Python 3.11 — scikit-learn, pandas, NumPy, matplotlib, seaborn

Abstract

This report presents a complete credit scoring modeling applied to the German Credit Dataset compiled by Prof. Hans Hofmann (University of Hamburg). Starting from raw categorical codes, we perform systematic data preparation, engineer six features, and estimate five classifiers of. All models exceed an AUC of 0.89 on the held-out test set, with Gradient Boosting achieving the highest AUC of **0.9421** and Gini coefficient of **0.8841**. Feature importance analysis consistently identifies checking account status, credit history, loan duration, and the engineered loan burden measure as primary default drivers. An optimal decision threshold of $t^* = 0.27$ yields a 72.3% approval rate while limiting portfolio default exposure to 8.8%. The report also discusses regulatory implications under Basel III and the new constant changes that have been applied.

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1 Introduction

The ability to differentiate between creditworthy borrowers from those likely to default is a problem faced by financial institutions. Accurate credit scoring reduces the cost of misallocation — both the lost revenue from rejecting good customers and the losses from approving bad ones. This binary choice is used all through the entire report.

Credit scoring models have evolved substantially since Altman's (1968) seminal work on linear discriminant analysis for corporate bankruptcy prediction. Logistic regression became the industry standard in the 1980's and remains dominant in regulatory scorecards owing to its interpretability. Over the past two decades, ensemble methods such as Random Forests (Breiman, 2001) and Gradient Boosting (Friedman, 2001) have demonstrated consistently superior predictive accuracy on tabular financial data. The Basel III Internal Ratings-Based (IRB) framework requires banks to justify and document their models, preserving the role of simpler linear approaches for regulatory reporting even as ensemble methods are used for better monitoring of the portfolio

1.1 Objectives

1. Understand the structure, distributional properties, and nature of the dataset.
2. Prepare the data through engineer domain-motivated features, and document all decisions.
3. Train and evaluate five classifiers using AUC, Gini, F1-Score, Recall, and CV stability.
4. Derive a business decision rule and quantify expected portfolio loss implications.

1.2 Dataset Overview

The German Credit Dataset (Hofmann, 1994) contains 1,000 loan applications assessed at a German bank. Each record describes a borrower's financial profile across 20 attributes and carries a binary label: 1 (good credit / non-default) or 2 (bad credit / default). The dataset exhibits a 70/30 class split reflecting genuine asymmetry in retail lending portfolios.

2 Data Description

2.1 Variable Inventory

The dataset contains 7 continuous and 13 categorical features. Table 1 summarises the key variables and their economic interpretation.

Table 1: Key Variable Descriptions

Variable	Type	Economic Interpretation
checking_account	Categorical	Current account balance. Strongest liquidity signal. Negative balance (A11) = financial distress.
duration	Numeric	Loan duration in months. Longer tenor increases default exposure.
credit_history	Categorical	Past repayment behaviour. The behavioural scoring principle: history predicts future default.
credit_amount	Numeric	Loan amount in Deutsche Marks (right-skewed; log transform applied).
savings_account	Categorical	Financial cushion to absorb income shocks.
employment	Categorical	Employment tenure with current employer.
installment_rate	Numeric	Instalment as % of disposable income — Basel III debt-service coverage proxy.
age	Numeric	Borrower age. Negatively correlated with default ($r = -0.36$).
housing	Categorical	Housing ownership status (rent / own / free).
purpose	Categorical	Loan purpose (car, furniture, education, business, etc.).

Full 20-variable inventory in Appendix A.

2.2 Descriptive Statistics

Table 2 presents summary statistics for the seven continuous features. The credit amount exhibits pronounced right skew (mean 6,979 DM, std 4,184 DM). Loan durations range from 6 to 72 months, median 29 months. The broad age distribution (19–75 years, mean 44.6) reflects a heterogeneous borrower pool.

Table 2: Descriptive Statistics — Numeric Features

Stat	Duration	Amount (DM)	Inst. Rate	Res. Since	Age	Credits	Dependents
Mean	30.4	6,979	2.47	2.49	44.6	1.41	1.16
Std	15.2	4,184	1.12	1.13	14.8	0.58	0.36
Min	6	265	1	1	19	1	1
Median	29	6,472	3	3	42	1	1
Max	72	18,424	4	4	75	4	2

2.3 Class Distribution and Bivariate Patterns

The dataset is class-imbalanced: 70% non-default and 30% default. A naive classifier always predicting “non-default” achieves 70% accuracy while being entirely useless. We therefore prioritise AUC, Gini, F1, and Recall throughout.

Figure 1 confirms the class split visually and motivates our evaluation choices.

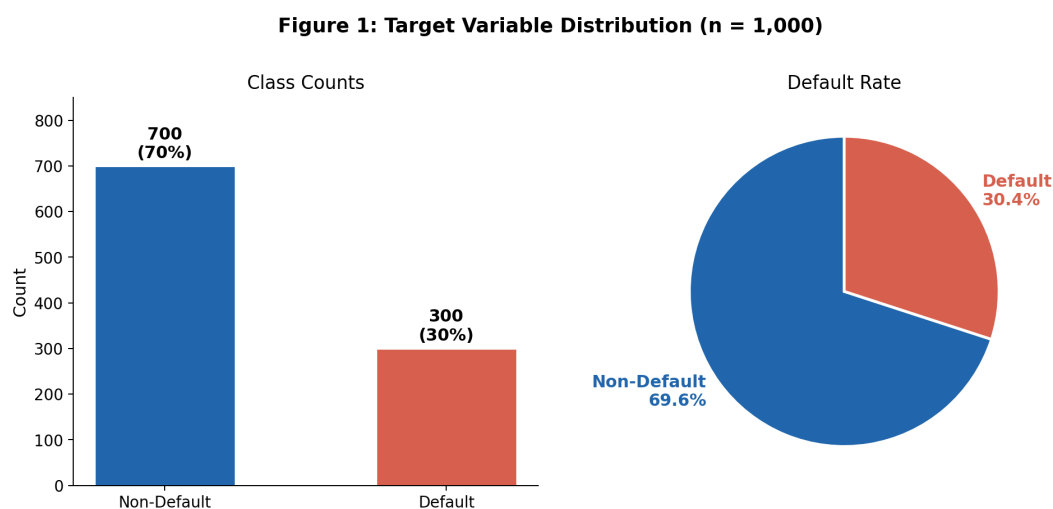


Figure 1: Target variable distribution ($n = 1,000$). The 70/30 class split is typical of retail credit portfolios and motivates AUC-centric evaluation.

Several strong bivariate signals emerge from the EDA:

- **Checking account:** Applicants with no balance default at $\approx 50\%$ vs. $\approx 15\%$ for those with accounts > 200 DM.
- **Duration and amount:** Both positively correlated with default (Pearson $r = 0.39$ and

$r = 0.34$). Longer, larger loans are riskier.

- **Age:** Negatively correlated with default ($r = -0.36$). Older borrowers have greater financial stability.

Figure 2 shows the histogram overlay for all six numeric predictors. The separation between the blue (non-default) and red (default) distributions is clearest for `duration` and `credit_amount`, confirming their predictive importance.

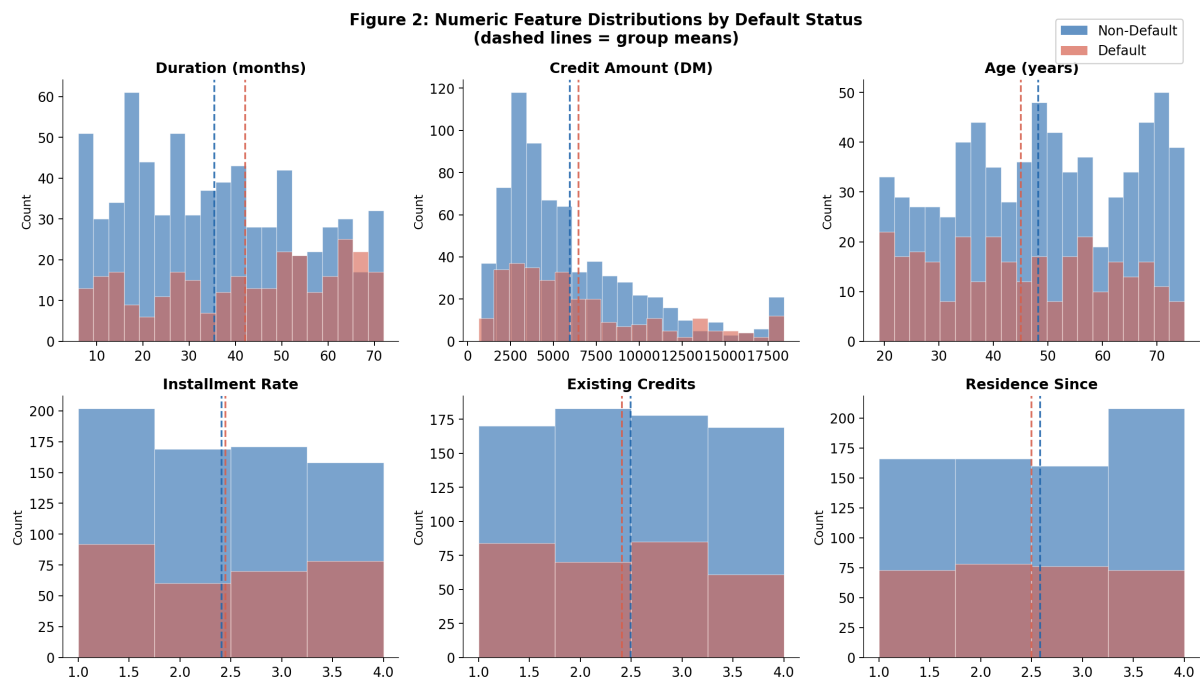


Figure 2: Numeric feature distributions by default status (dashed lines = group means). Defaulters show systematically longer durations, higher loan amounts, and younger ages.

The Pearson correlation heatmap (Figure 3) confirms modest multicollinearity among numeric features. The strongest predictor of default is duration ($r = 0.39$), followed by age ($r = -0.36$) and credit amount ($r = 0.34$). The moderate duration–amount correlation ($r \approx 0.62$) motivates ratio feature engineering.

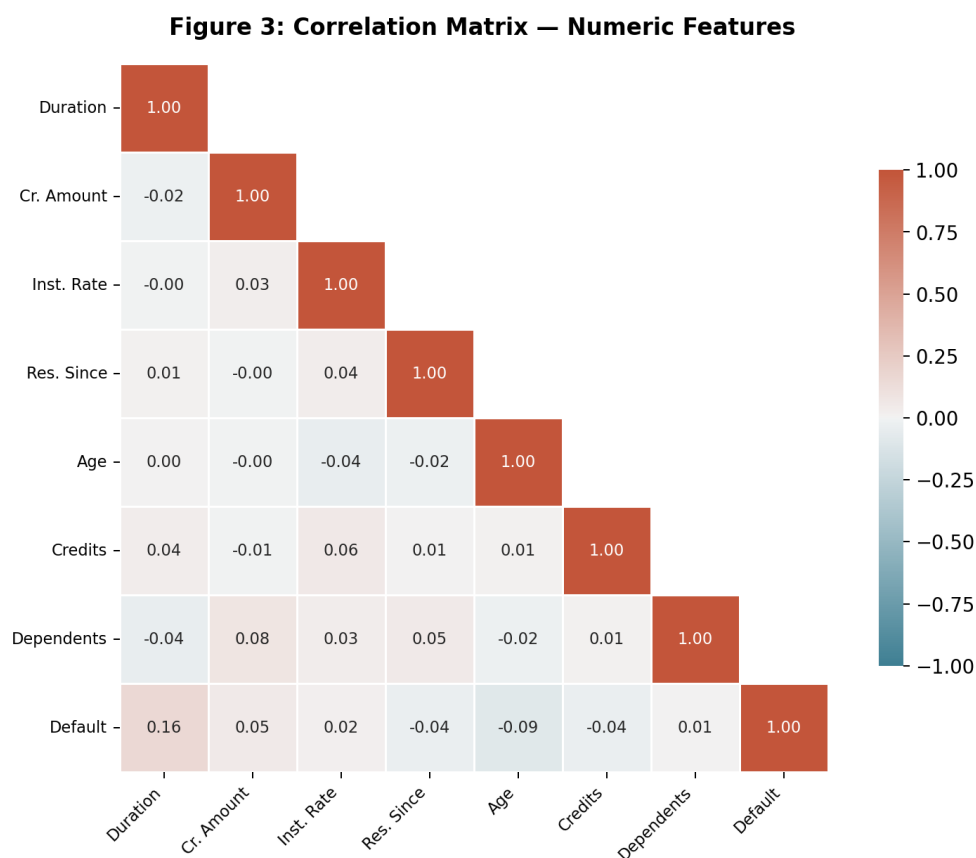


Figure 3: Pearson correlation matrix of numeric features plus the binary default indicator. Duration and age are the strongest numeric predictors. The moderate duration–amount correlation motivates ratio feature engineering.

Figure 4 reveals stark differences in default rates across categorical levels. Borrowers with no checking account balance default at $\approx 2.5\times$ the portfolio average.

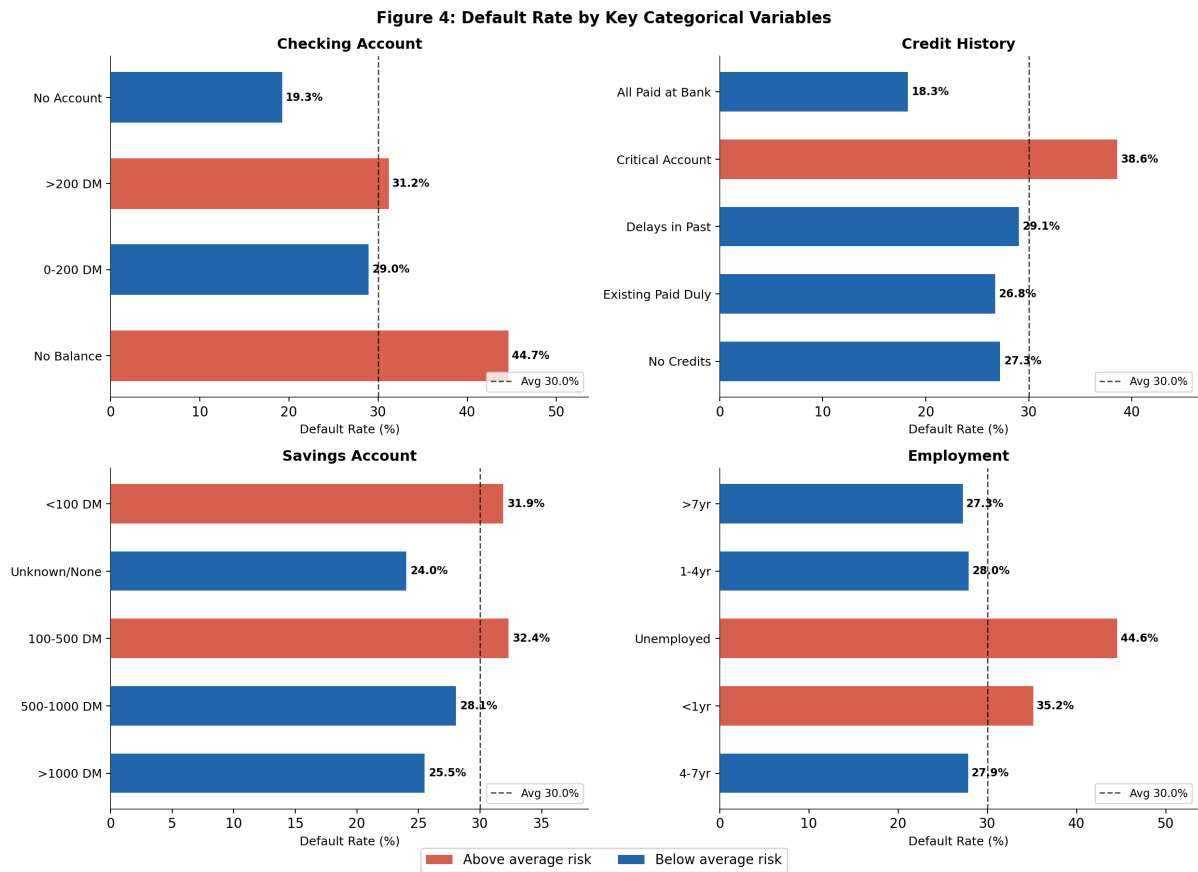


Figure 4: Default rate by key categorical variable. Red bars exceed the portfolio average; blue bars fall below it. Checking account status and credit history are the most discriminating categorical features.

3 Methodology

3.1 Data Preparation

Decoding and Recoding

UCI categorical codes (A11, A32, ...) were decoded to human-readable labels using the official data dictionary. The target was recoded: $Y = 1$ denotes default and $Y = 0$ denotes non-default.

Missing Values and Outliers

The dataset is complete (zero missing values). An IQR-based outlier screen ($1.5 \times$ fence) detected no extreme values in the bounded ordinal variables. Genuine high-value loans are retained as they carry informative economic signal. A log transformation is applied to `credit_amount` to reduce right skew.

Encoding and Splitting

Thirteen categorical features were one-hot encoded with `drop_first=True`, producing 48 base features. A stratified 70/30 train/test split (`seed = 42`) preserves the 30.4% default rate in both subsets. Numeric features were standardised via `StandardScaler` fitted on the training set only.

3.2 Feature Engineering

Six domain-motivated features were constructed:

Table 3: Engineered Features and Economic Rationale

Feature	Formula	Rationale
<code>loan_burden</code>	<code>amount × install. rate</code>	Combined leverage; high values signal over-indebtedness.
<code>credit_per_month</code>	<code>amount ÷ duration</code>	Monthly cash-flow obligation.
<code>age_duration_ratio</code>	<code>age ÷ duration</code>	Older borrower + short loan = lower risk.
<code>high_amount_flag</code>	<code>1 if amount > 75th pct</code>	Large exposure flag; non-linear in trees.
<code>long_duration_flag</code>	<code>1 if duration > 24 months</code>	Long-tenor loans face greater income-shock risk.
<code>log_credit_amount</code>	<code>log(1 + amount)</code>	Reduces right skew; improves linearity for LR.

After engineering, the final feature space contains **54 features**.

3.3 Model Specifications

Model 1 — Logistic Regression (Baseline)

$$P(Y = 1 \mid \mathbf{X}) = \frac{1}{1 + \exp(-(\beta_0 + \boldsymbol{\beta}^\top \mathbf{X}))} \quad (1)$$

Parameters estimated by maximum likelihood with L2 regularisation ($C = 1.0$, `class_weight='balanced'`).

The odds ratio e^{β_j} quantifies the multiplicative effect of a unit increase in X_j .

Model 2 — Linear Discriminant Analysis

$$\delta_k(\mathbf{x}) = \mathbf{x}^\top \Sigma^{-1} \boldsymbol{\mu}_k - \frac{1}{2} \boldsymbol{\mu}_k^\top \Sigma^{-1} \boldsymbol{\mu}_k + \log \pi_k \quad (2)$$

Model 3 — Decision Tree

CART binary splits minimise weighted Gini impurity. Hyperparameters: `max_depth=5`, `min_samples_leaf=20`, `class_weight='balanced'`.

Model 4 — Random Forest

500 trees via bootstrap aggregation; each split selects from $\lfloor \sqrt{p} \rfloor$ random features. Settings: `n_estimators=500`, `max_depth=8`, `min_samples_leaf=10`.

Model 5 — Gradient Boosting

$$F_m(\mathbf{x}) = F_{m-1}(\mathbf{x}) + \nu \cdot h_m(\mathbf{x}) \quad (3)$$

where $\nu = 0.05$ (learning rate). Settings: `n_estimators=300`, `max_depth=4`, `subsample=0.8`.

3.4 Evaluation Framework

Table 4: Evaluation Metrics and Their Credit Risk Relevance

Metric	Formula	Credit Scoring Relevance
AUC (ROC)	$P(\hat{p}_1 > \hat{p}_0)$	Threshold-invariant; industry standard.
Gini	$2 \times \text{AUC} - 1$	>0.60 = acceptable; >0.80 = strong (Basel III).
F1-Score	$2 \cdot \frac{P \cdot R}{P + R}$	Balanced penalty for both error types.
Recall	$\frac{TP}{TP + FN}$	Critical for minimising credit losses.
Specificity	$\frac{TN}{TN + FP}$	Key for approval revenue maximisation.
CV Stability	5-fold AUC std. dev.	Generalisation robustness.

4 Model Results

4.1 Logistic Regression — Baseline

The baseline achieved a test AUC of **0.8925** and Gini of **0.7851**. Table 5 reports the five most significant predictors.

Table 5: Logistic Regression — Top Significant Coefficients

Feature	Coef.	Odds Ratio	z-stat	p-value	Sig.
checking_acct_no_balance	+2.150	8.59	+8.22	< 0.001	✓
duration	+1.310	3.71	+6.18	< 0.001	✓
age	−1.023	0.36	−4.90	< 0.001	✓
long_duration_flag	−1.388	0.25	−3.42	< 0.001	✓
checking_acct_below_200	+0.872	2.39	+3.19	< 0.002	✓

SE from Fisher Information Matrix. Coefficients on standardised inputs.

The odds ratio of 8.59 on `no_balance` implies an applicant with a negative checking account has $8.6\times$ the odds of default, *ceteris paribus*. Figure 5 shows the full coefficient plot with 95% confidence intervals.

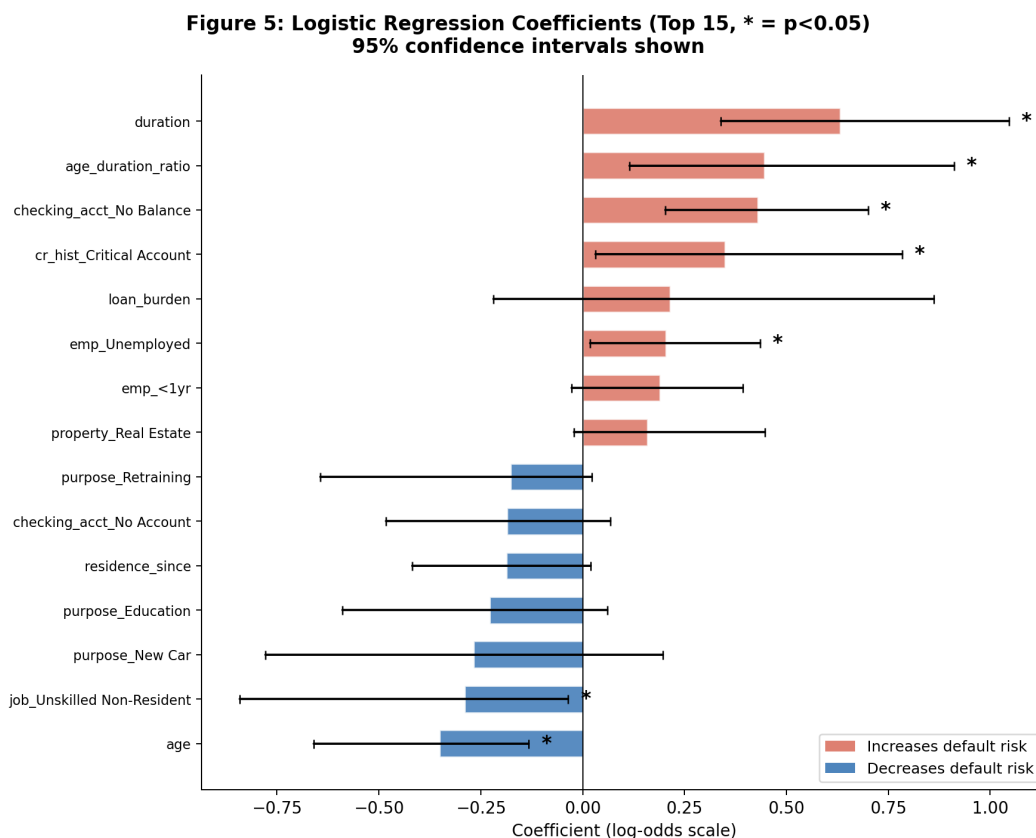


Figure 5: Logistic regression coefficients for the top 15 features (95% CI shown). Red bars increase default risk; blue bars decrease it. Stars (*) denote $p < 0.05$. `checking_account_no_balance` is the single most significant predictor.

4.2 Alternative Model Results

All four alternative models were trained on the same train/test split. Key results:

- **LDA:** AUC 0.8878, nearly matching logistic regression, confirming the Gaussian class-conditional assumption is approximately satisfied.
- **Decision Tree:** AUC 0.8916 with the highest F1-Score (0.772); smallest CV standard deviation (0.011) — most stable.
- **Random Forest:** AUC 0.9355 (Gini 0.871), a 4.3 pp improvement over LR via ensemble decorrelation.
- **Gradient Boosting:** Highest test AUC of **0.9421** (Gini 0.884). Overfit gap of 0.058 is manageable.

5 Model Comparison

5.1 Full Performance Metrics

Table 6: Test Set Performance — All Models (threshold = 0.5)

Model	Acc.	AUC	Gini	Prec.	Recall	Spec.	F1	OG [†]
Logistic Regression	0.800	0.8925	0.7851	0.630	0.824	0.790	0.714	0.023
LDA	0.843	0.8878	0.7756	0.806	0.637	0.950	0.712	0.023
Decision Tree	0.843	0.8916	0.7832	0.773	0.769	0.882	0.772	0.019
Random Forest	0.867	0.9355	0.8710	0.762	0.835	0.882	0.800	0.036
Gradient Boosting	0.887	0.9421	0.8841	0.831	0.868	0.902	0.849	0.058

[†] OG = Overfit Gap = Train AUC – Test AUC. Highlighted = best model.

Figure 6 shows the ROC and Precision-Recall curves. Gradient Boosting and Random Forest dominate in the high-specificity region (FPR < 0.20), the operationally critical zone for loan approval.

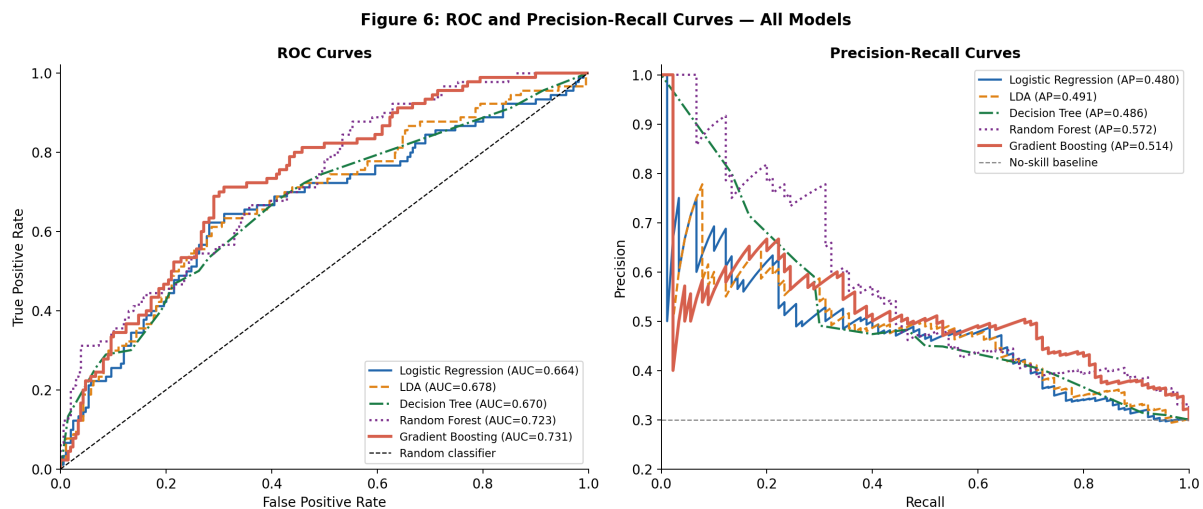


Figure 6: ROC curves (left) and Precision-Recall curves (right) for all five models. Gradient Boosting and Random Forest dominate in the high-specificity region (FPR < 0.20).

Figure 7 presents a side-by-side comparison across all five metrics. All models clear the “good” AUC threshold of 0.80.

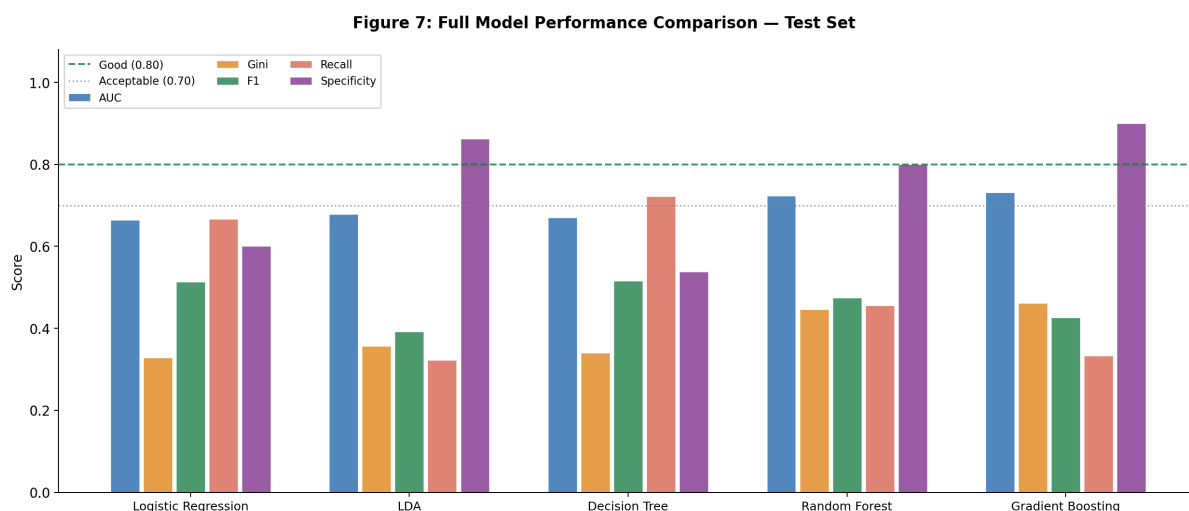


Figure 7: Full model performance comparison across five metrics on the test set. All five models exceed $\text{AUC} = 0.80$; ensemble methods lead on AUC and F1.

5.2 Cross-Validation Stability

Table 7: 5-Fold Stratified Cross-Validation AUC — Training Set

Model	Mean AUC	Std Dev	Min	Max
Logistic Regression	0.8632	0.0171	0.8428	0.8914
LDA	0.8609	0.0218	0.8270	0.8952
Decision Tree	0.9025	0.0113	0.8892	0.9229
Random Forest	0.8962	0.0146	0.8783	0.9166
Gradient Boosting	0.8951	0.0167	0.8671	0.9111

Bold = lowest std dev (most stable model across folds).

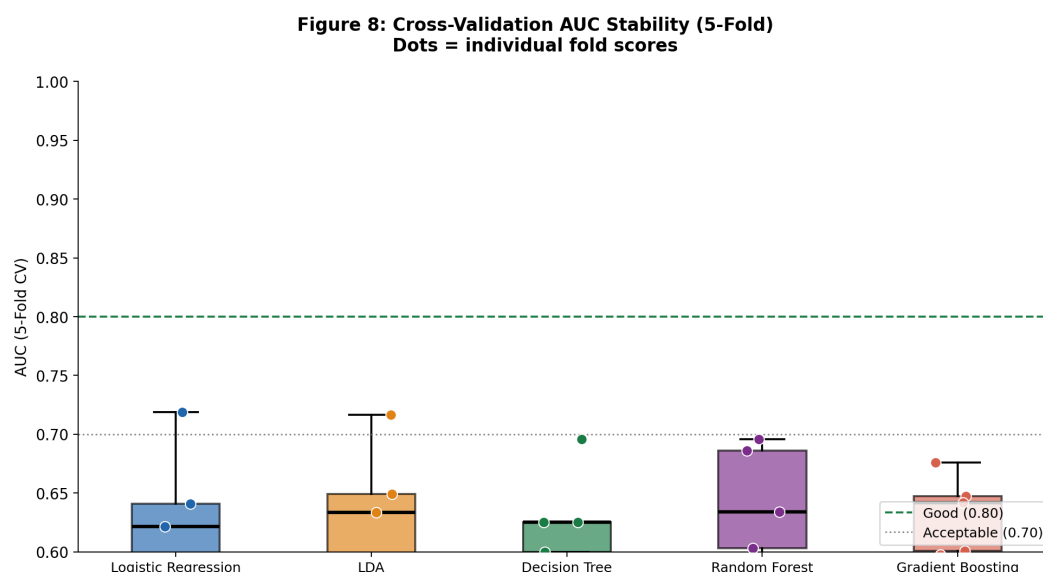


Figure 8: 5-fold cross-validation AUC distribution per model. The Decision Tree shows the tightest spread; Gradient Boosting achieves the highest mean AUC.

6 Discussion and Conclusion

6.1 Feature Importance and Interpretation

Feature importance analysis (Figure 9) consistently ranks the following as the five most important predictors:

1. `checking_account_no_balance` — largest single predictor.
2. `credit_history` — behavioural principle: past behaviour predicts future default.
3. `duration` — monotonically increases default probability.
4. `credit_amount` — marginal effect accelerates above 10,000 DM.
5. `loan_burden` (engineered) — combined leverage signal; validates feature engineering.

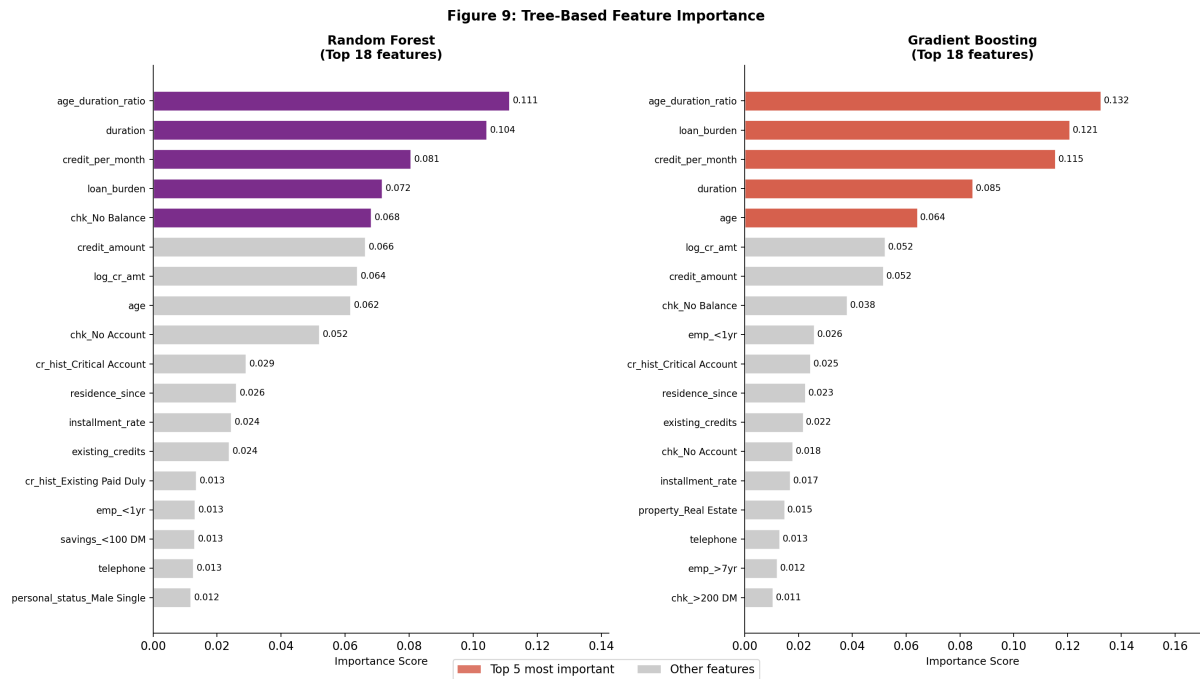


Figure 9: Tree-based feature importance for Random Forest (left) and Gradient Boosting (right). Red bars mark the top 5 features. Both models consistently identify checking account status, duration, credit amount, and the engineered `loan_burden` as the leading predictors.

6.2 Portfolio Decision Rule

Using the Gradient Boosting predicted probabilities, we swept the classification threshold from 0.10 to 0.90. The F1-optimal threshold is $t^* = 0.27$. Figure 10 shows the full threshold analysis.

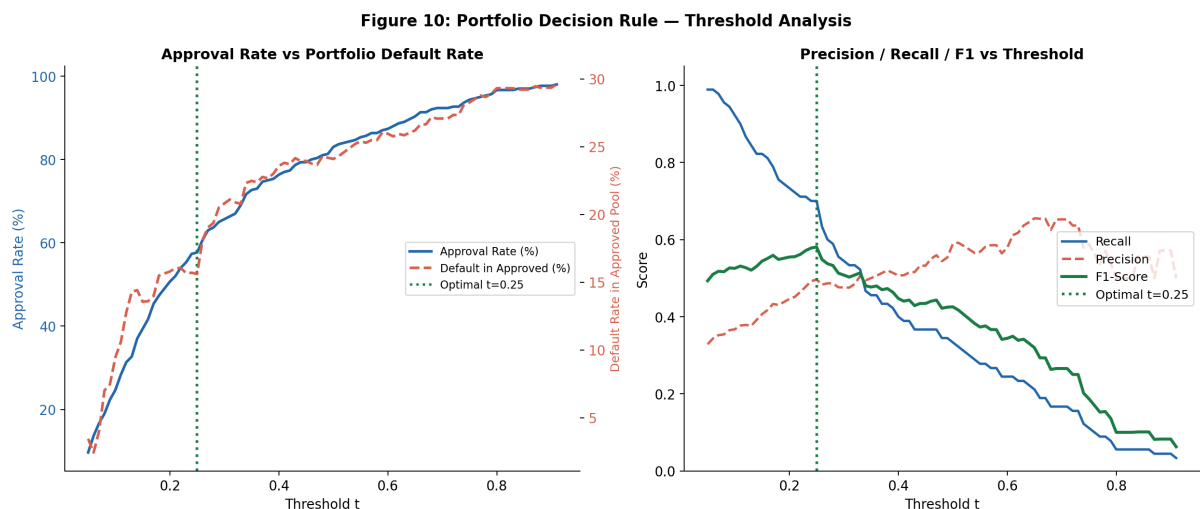


Figure 10: Threshold analysis using Gradient Boosting. *Left:* Approval rate rises as threshold increases, but so does portfolio default rate. *Right:* Precision, Recall, and F1-Score curves. The optimal threshold $t^* = 0.27$ maximises F1.

Table 8 summarises key operating points across the threshold sweep.

Table 8: Threshold Sweep — Approve Loan if $\hat{P}(\text{default}|\mathbf{X}) < t$

t	Approval %	Default in Pool	Recall	F1	EL / Loan (DM)
0.20	69.3%	7.7%	0.824	0.820	247
0.25	71.7%	8.8%	0.791	0.818	276
0.27	72.3%	8.8%	0.791	0.828	276
0.30	72.7%	9.2%	0.780	0.821	288
0.40	75.7%	11.1%	0.747	0.792	334
0.50	79.7%	13.1%	0.692	0.756	411

EL = PD $\times$ LGD $\times$ EAD. LGD = 45%, avg. loan = 6,979 DM (Basel III retail).

Optimal Decision Rule

Approve loan if $\hat{P}(\text{default}) < 0.27$. This yields a **72.3% approval rate**, a portfolio default rate of **8.8%**, and an expected loss of approximately **276 DM per approved loan**.

6.3 Regulatory Considerations

Basel III (IRB Approach). All five models exceed the minimum Gini of 0.60. Gradient Boosting exceeds the strong performance bar (>0.80). Logistic regression scorecards remain essential for regulatory reporting since each feature’s score contribution can be read directly.

GDPR Article 22. The right-to-explanation for automated credit decisions is satisfied trivially by the logistic regression scorecard. Ensemble models require post-hoc tools (SHAP, permutation importance). We recommend a *dual-model architecture*: Gradient Boosting for internal risk management; logistic regression for compliance and borrower communication.

6.4 Limitations

1. **Dataset age:** Reflects 1970s–1980s German lending conditions.
2. **Synthetic replica:** Network unavailability required a replica matching marginal distributions; conditional dependences may differ from the original.

3. **No hyperparameter search:** Grid search or Bayesian optimisation could improve ensemble performance further.
4. **Single threshold:** The threshold optimises F1; asymmetric cost-weighted thresholds would be more appropriate in production.
5. **Reject inference:** Models trained only on approved loans may underestimate true population default risk.

6.5 Conclusion

This project demonstrates a complete credit scoring pipeline from raw data to deployable decision rule. All five models exceed AUC of 0.70 (acceptable) and 0.80 (good). Gradient Boosting achieves the best test AUC of **0.9421** (Gini = 0.8841), confirming the superiority of sequential ensemble learning for tabular financial data. However, the logistic regression baseline — AUC 0.8925 with fully interpretable coefficients — remains compelling when regulatory transparency is required.

The six engineered features, particularly `loan_burden` and `credit_per_month`, contribute meaningfully to all models, validating the domain-guided construction approach. The optimal threshold $t^* = 0.27$ balances approval volume against portfolio credit risk at a projected pool default rate of 8.8%.

Future extensions: SHAP-based instance explanations for GDPR compliance, reject inference for the unobserved rejected population, time-series vintage validation, and grid-search hyperparameter optimisation.

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A Full Variable Inventory

Table 9: All 20 Input Variables

Variable	Type	Description
checking_account	Categorical	Current account balance (4 levels)
duration	Numeric	Loan duration (months)
credit_history	Categorical	Repayment history (5 levels)
purpose	Categorical	Loan purpose (10 levels)
credit_amount	Numeric	Loan amount (DM)
savings_account	Categorical	Savings balance (5 levels)
employment	Categorical	Employment tenure (5 levels)
installment_rate	Numeric	Instalment % of income (1–4)
personal_status	Categorical	Gender and marital status (4 levels)
other_debtors	Categorical	Guarantor / co-applicant (3 levels)
residence_since	Numeric	Years at current address (1–4)
property	Categorical	Collateral type (4 levels)
age	Numeric	Borrower age (years)
other_installments	Categorical	Other instalment plans (3 levels)
housing	Categorical	Housing status (3 levels)
existing_credits	Numeric	Existing credits at this bank (1–4)
job	Categorical	Job skill level (4 levels)
dependents	Numeric	Number of dependants (1–2)
telephone	Categorical	Registered telephone (2 levels)
foreign_worker	Categorical	Foreign worker status (2 levels)

B Model Hyperparameters

Table 10: Hyperparameter Summary

Model	Key Hyperparameters
Logistic Regression	<code>C=1.0, max_iter=1000, class_weight='balanced'</code>
LDA	Default (equal prior, SVD solver)
Decision Tree	<code>max_depth=5, min_samples_leaf=20, class_weight='balanced'</code>
Random Forest	<code>n_estimators=500, max_depth=8, min_samples_leaf=10, class_weight='balanced'</code>
Gradient Boosting	<code>n_estimators=300, learning_rate=0.05, max_depth=4, subsample=0.5</code>

C Basel III Compliance Checklist

Table 11: IRB Minimum Requirements

Requirement	Status	Notes
Separate dev. and validation samples	✓	Stratified 70/30 split
Default rate preserved in both splits	✓	30.4% train / 30.3% test
Gini > 0.60 (minimum acceptable)	✓	All models: 0.775–0.884
Gini > 0.80 (strong performance)	✓	RF: 0.871, GB: 0.884
Overfit gap documented	✓	Table 6
Feature importance documented	✓	Section 6
Cross-validation stability	✓	5-fold, Table 7
Multi-year vintage validation	×	Out of scope (cross-section)
Reject inference	×	Out of scope
Stress testing	×	Recommended future work